1. Overview

The existing workflow was reviewed and manually tested node by node. Testing covered data mapping, conditional routing, AI-generated lesson content, document creation and email delivery. A test-data end-to-end execution was also performed.

The main functional issue was that the original Google Docs route displayed Markdown syntax as plain text. A new path was tested using Markdown-to-HTML conversion, a JavaScript multipart request body, and a Google Drive HTTP upload. The resulting document displayed formatted content. Gmail was updated to use the new document link.

The Append Row node was also tested separately using a QA sheet; the tester confirmed that a row was added. It was disabled in the client handoff template to avoid unnecessary writes or duplicated records.

2. Test scope and results

The following statuses reflect observed manual testing, the tester's confirmations, and configuration review. "PASS by tester" indicates user-confirmed behavior rather than an independently captured output. Some AI content-quality issues and live-deployment checks remain open.

Test case / Component	Status	Observed result / Note
TC-1 Google Sheets Trigger (Fetch Test Event)	PASS	34 sample records; live new-row activation not proven.
TC-2 Edit Fields	PASS	Six required fields mapped for one input item.
TC-3 If - valid route	PASS	Complete input routed to true branch.
TC-4 AI-Lesson outline	PASS	45-minute lesson outline generated; scope concerns noted.
TC-5 Students Activities	PASS	Five activities generated; some advanced/out of scope.
TC-6 Merge1	PASS	Activities and outline retained in distinct fields.
TC-7 AI Assessment + retests	OPEN CONTENT ISSUE	Answer key removed; multiple-choice quality was variable across runs.
TC-8 Merge2	PASS	Assessment, activities, lesson outline preserved.
TC-9 Assessment Answer key	PASS	Numbered answers generated; upstream item ambiguities persist.
TC-10 Edit Fields2	PASS	Answer key mapped to separate answer_key field.
TC-11 Merge3	PASS	Four content fields retained in merged item.
TC-12 AI Differentiated material	PASS	Advanced material generated; some incorrect debugging instructions.
TC-13 Edit Fields1	PASS	Five sections assembled in Final_content.
TC-14 Create a document (legacy route)	PASS	Test Google Doc created; route later disconnected.
TC-15 Update a document (legacy route)	PASS	Update accepted; document body verified separately.
TC-16 Gmail Send (legacy route)	PASS	SENT message ID returned.
TC-17 Email link access (legacy route)	PASS by tester	Email received and link opened for test account.
TC-18 If - missing Topic	PASS	Blank Topic routed to false branch.
TC-19 Markdown to HTML	PASS	HTML field generated; five sections preserved (shared JSON).
TC-20 Full main-path integration	PASS	User screenshot showed green nodes through Gmail.

Test case / Component	Status	Observed result / Note
TC-21 Code - HTML multipart prep	PASS	uploadBody and contentType returned (shared JSON).
TC022 HTTP Request - HTML conversion	PASS	Google Docs file id, MIME type and URL returned (shared JSON).
TC023 Formatted Google Doc	PASS	Rendered headings, bold, lists; latest link reconfirmed.
TC024 Gmail link preview	PASS	Expression resolved latest HTTP Request webViewLink.
TC-25 Formatted-route wiring	PASS	HTTP Request connected to Gmail; legacy path detached.
TC-26 New Gmail route	PASS	Tester confirmed new path operates; no dedicated JSON screenshot.
TC-27 Missing-information Gmail	PASS	SENT label and message ID returned on invalid-input path.
TC-28 Append row (separate test sheet)	PASS	Mapped response added; tester confirmed row in test sheet.

3. Issues and remaining limitations

D-01 | Append-row duplication risk

A side-effecting append stage previously targeted the form data pipeline. It was safely tested on a separate QA sheet. Client template keeps the node disabled to avoid unintended writes.

D-02 | MCQ ambiguities / Python correctness

AI Assessment retests removed the duplicated answer key, but examples included valid alternatives in more than one option and uppercase AND. These errors are generated-content quality issues and remain a regression risk.

D-03 | Differentiation beyond lesson scope

Advanced output sometimes introduced loops, functions, exceptions and other techniques inconsistent with a 45-minute Python basics objective. Educational quality requires human review.

D-04 | Google Docs raw Markdown

Legacy Google Docs Update inserted Markdown markers (### and **). Corrected flow converts Markdown to HTML then uses Drive multipart upload/conversion. Visual formatting was checked.

D-05 | Fixed test resources and recipient

Source exported with tester Gmail address, source sheet, pinned row, test folder and credential references. Client template removes references and requires client-specific configuration.

D-06 | Live trigger and recipient access

Manual Fetch Test Event and pinned-input end-to-end tests do not prove unattended new-row triggering. Document link opened for the tester; access for other recipients still depends on Drive sharing.

4. Client setup required

The delivered importable JSON is a template with client-specific references removed. It was not re-executed after the placeholder values were inserted. Before activating the imported workflow, complete these steps:

1. Connect client-owned credentials. Google Sheets Trigger, Sheets, Google Drive, Gmail, and Gemini/AI model.

2. Select spreadsheet and response tab. Replace placeholder spreadsheet ID and tab ID in trigger; preserve column names/spaces.

3. Set output folder. Replace REPLACE_WITH_DESTINATION_FOLDER_ID in Code in JavaScript.

4. Set email recipients. Change CONFIGURE_RECIPIENT@example.com on both Gmail nodes; consider adding a teacher-email field.

5. Keep append node disabled. Separate append test passed, but appending is unnecessary for the main flow.

6. Verify document sharing. Check access as the actual recipient, not only as the connected Google account.

7. Run genuine end-to-end validation. Submit a new Form response with the trigger active and check document + email.

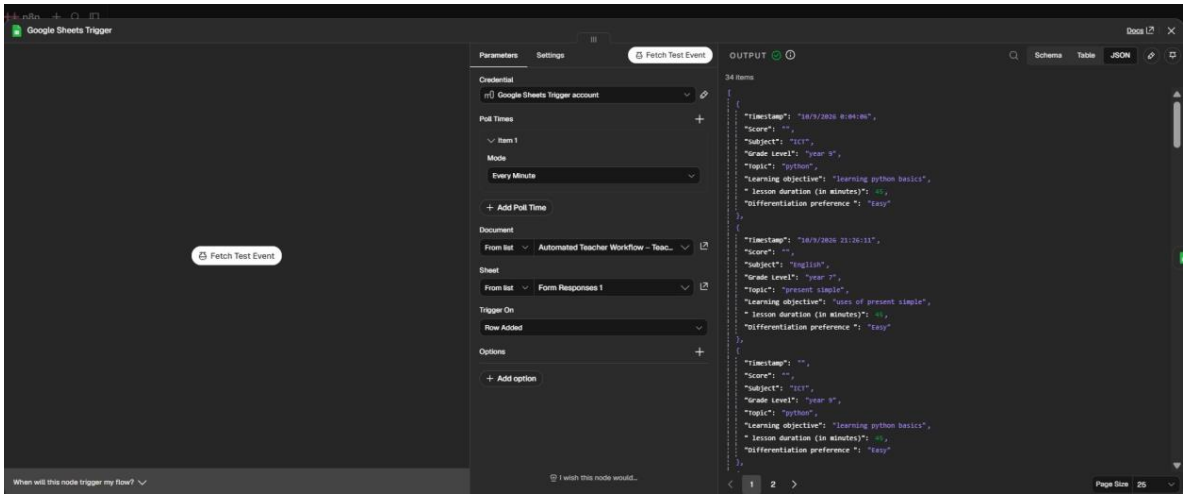
8. Inspect educational material. Review questions, answer key, and differentiated content for accuracy and age appropriateness.

Additional checks: keep the workflow inactive until the real trigger, Google Drive sharing permissions, valid inputs, missing-field handling and educational correctness have been confirmed in the client environment.

5. Screenshot evidence

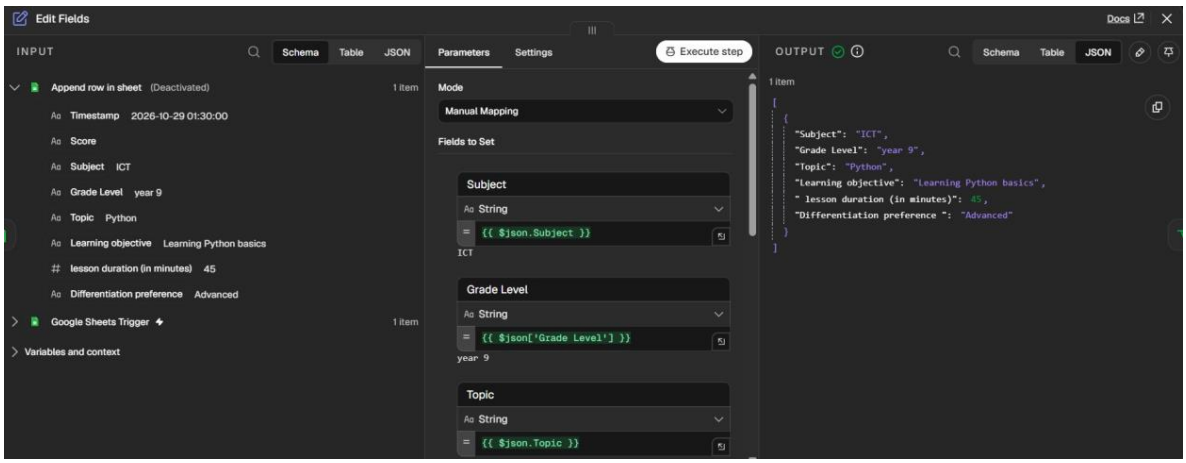
26 screenshots from the testing session are included below. The image files are also provided separately in the Evidence folder. Some email identifiers or document links have been redacted.

TC-1 - Google Sheets Trigger Fetch



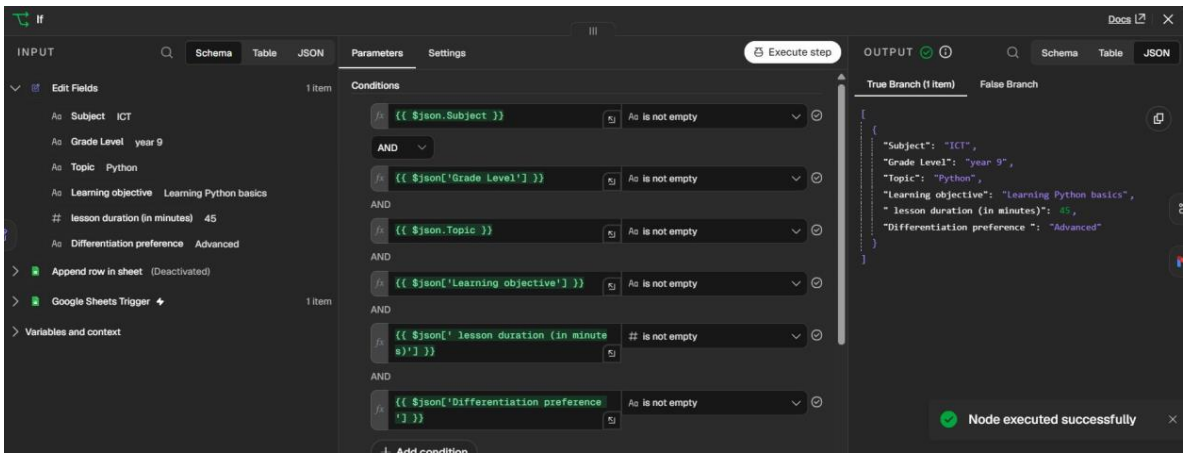
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-2 - Edit Fields Mapping



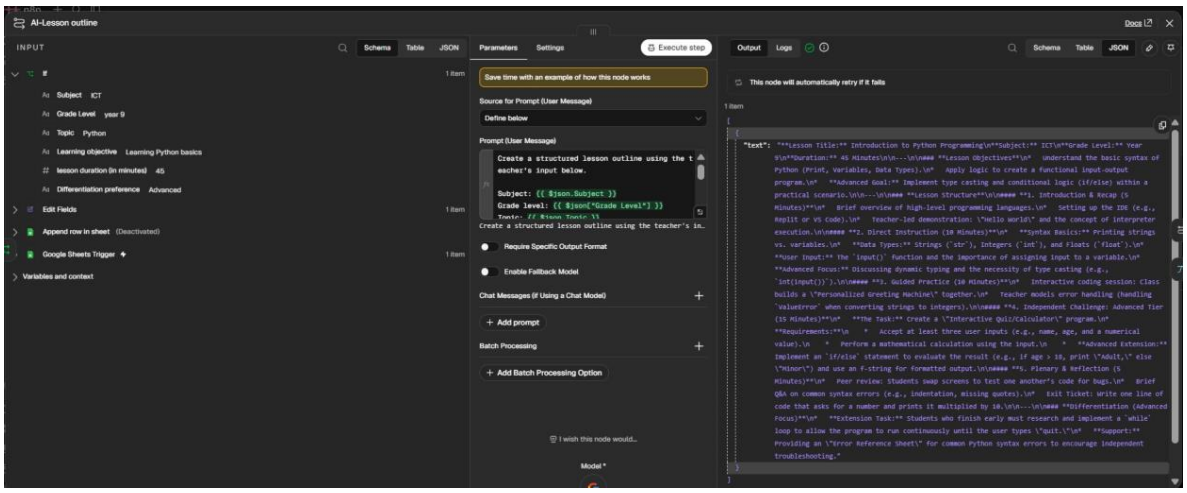
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-3 - If Valid True Branch



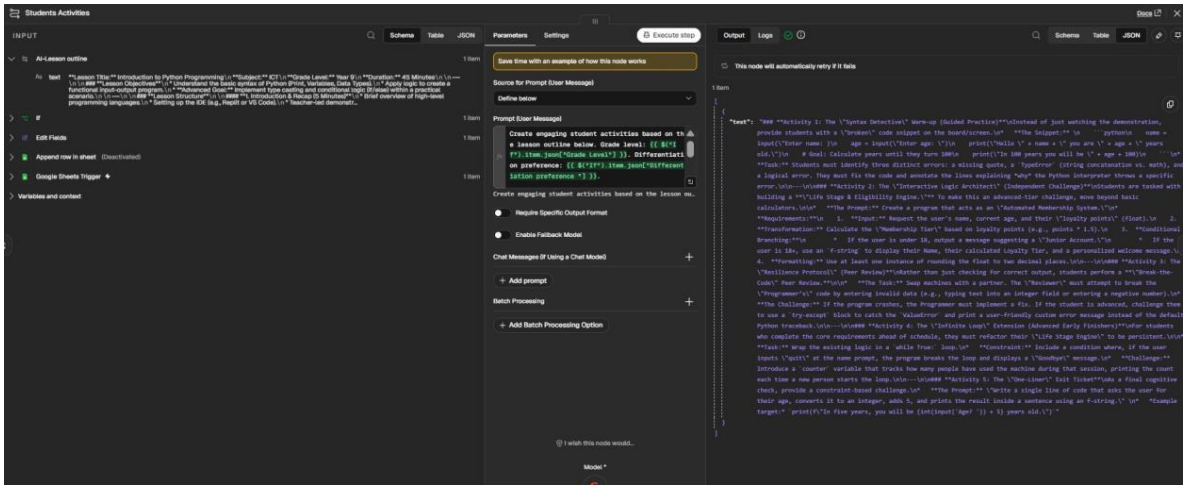
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-4 - AI Lesson Outline



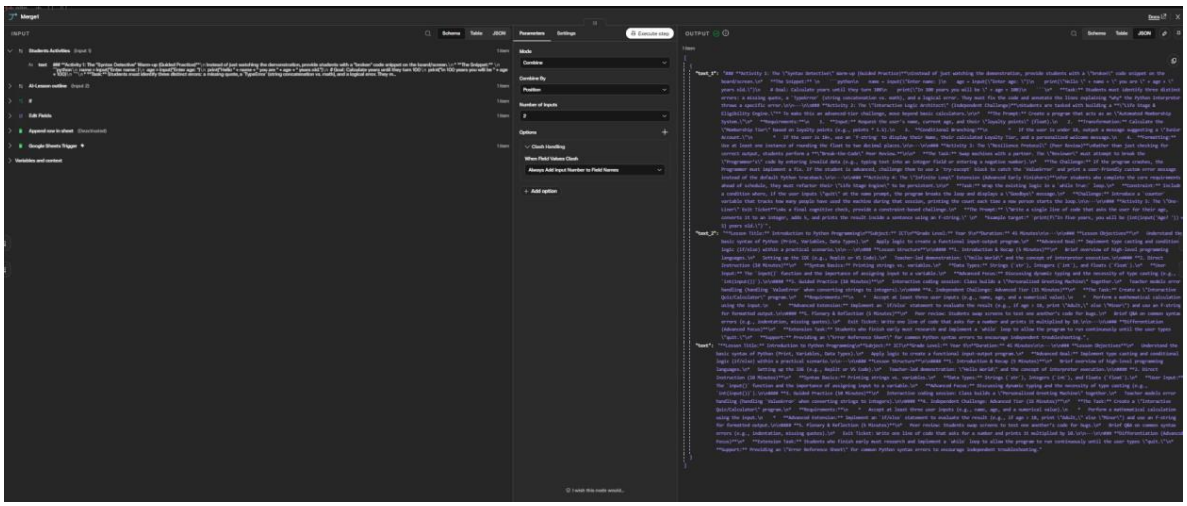
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-5 - Student Activities



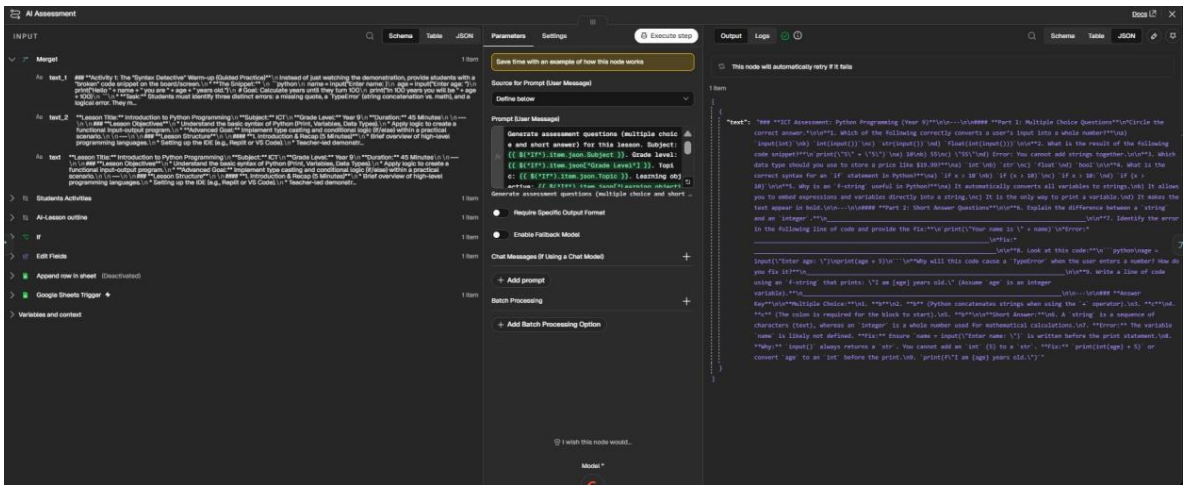
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-6 - Merge1



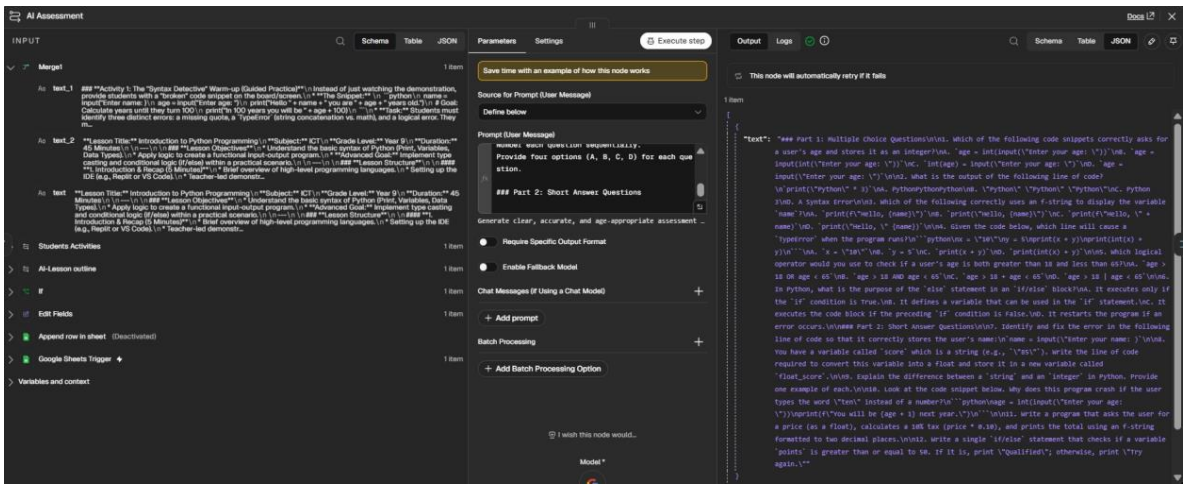
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-7 - AI Assessment Initial Problem



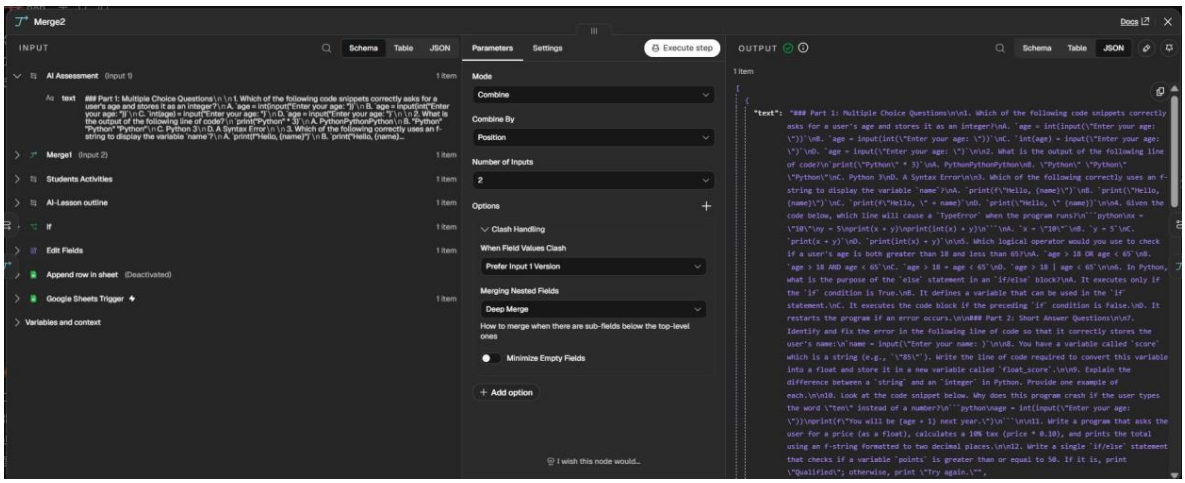
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-7 - AI Assessment Retest 3



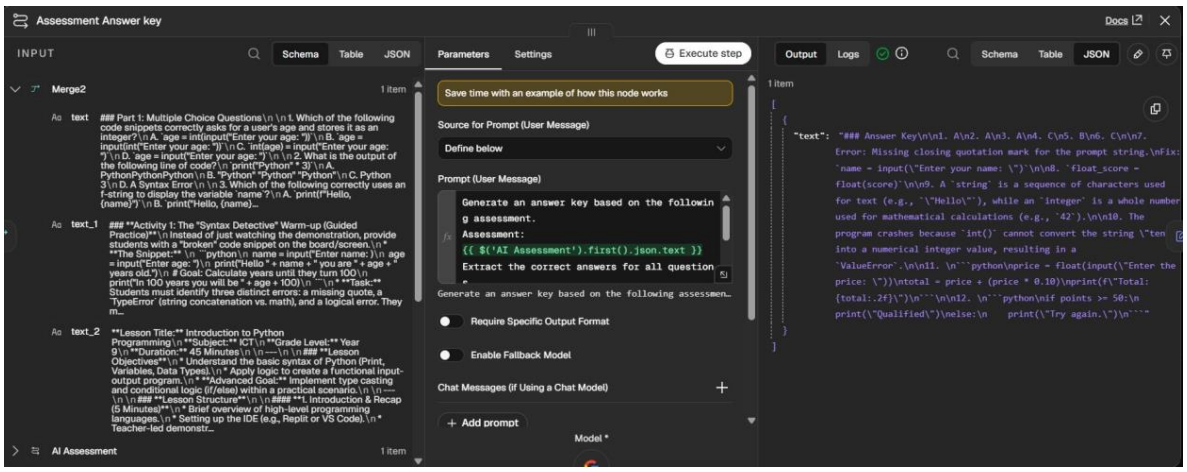
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-8 - Merge2



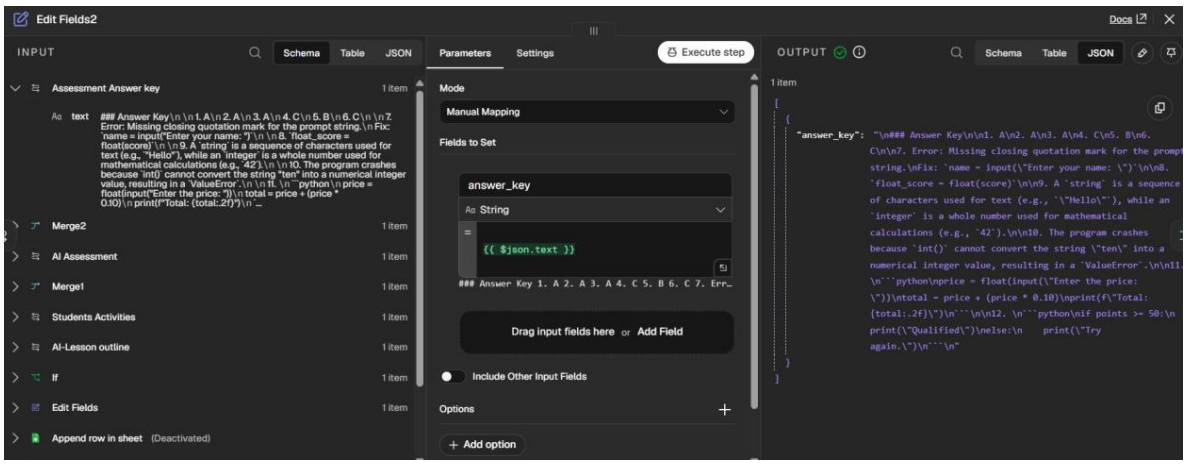
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-9 - Assessment Answer Key



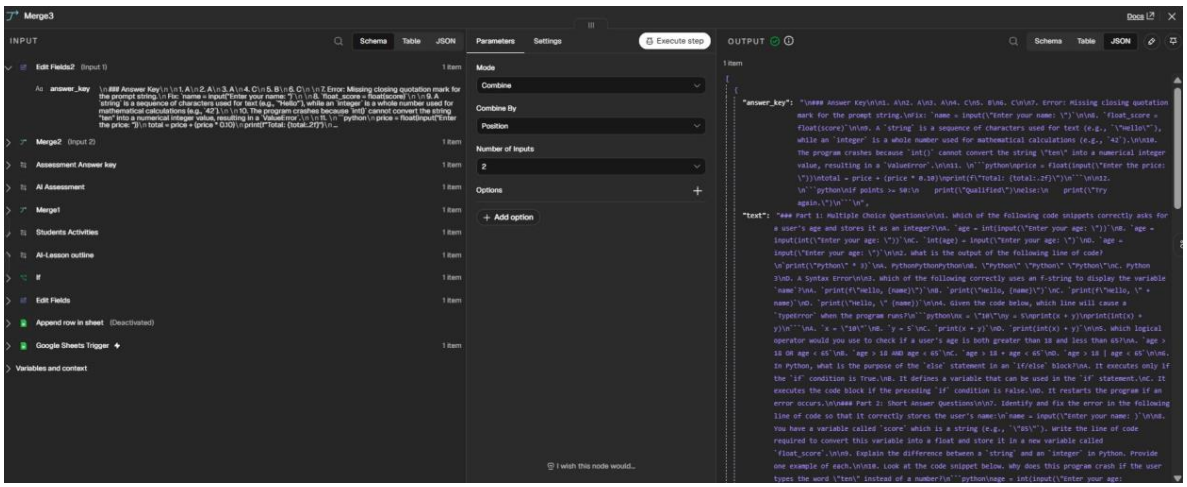
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-10 - Edit Fields2



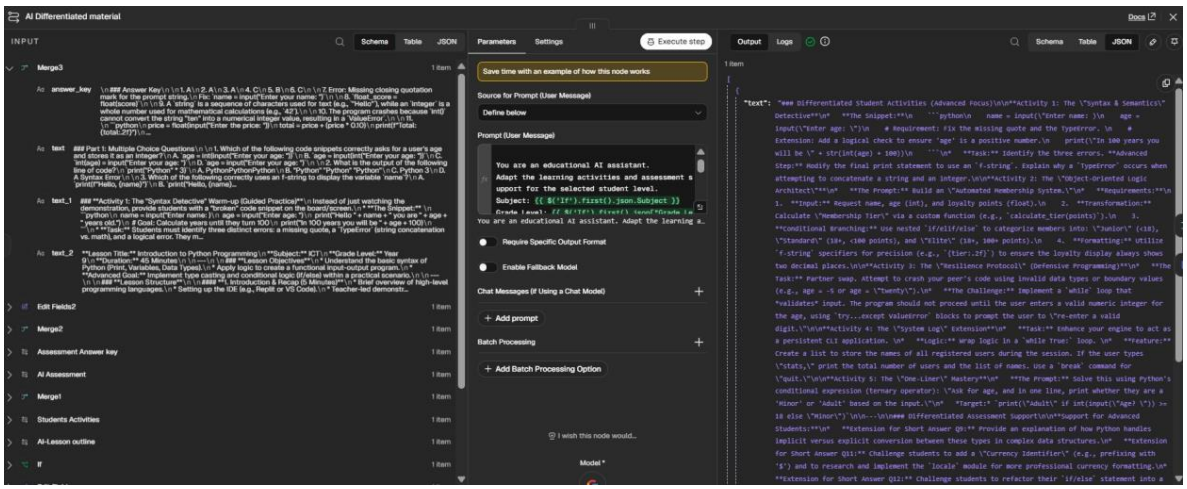
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-11 - Merge3



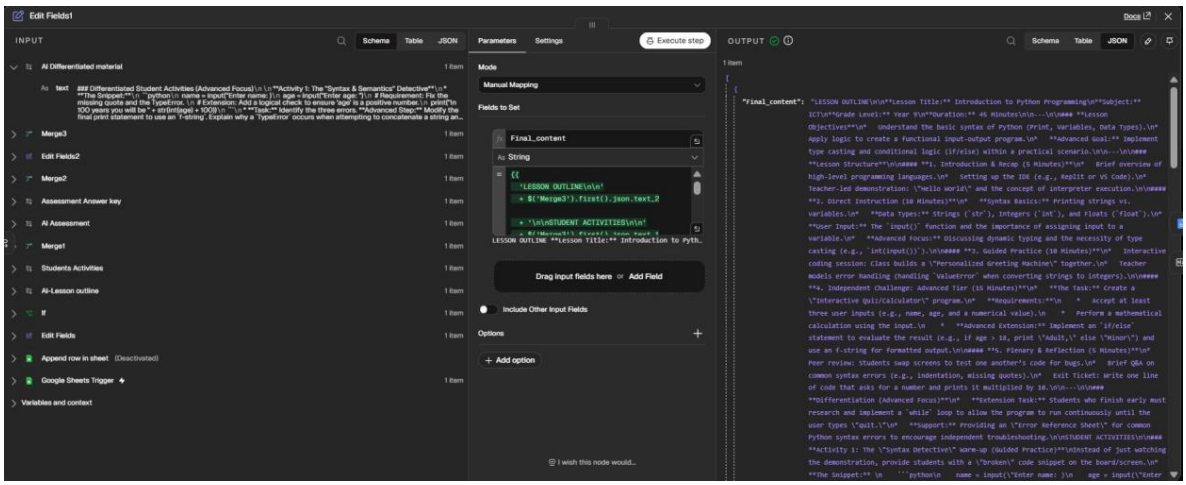
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-12 - Differentiated Material



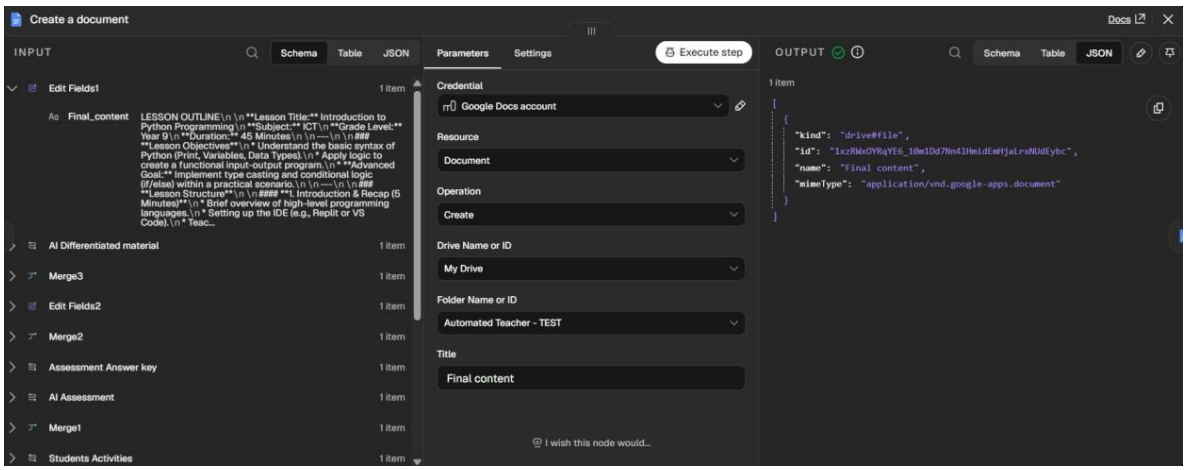
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-13 - Final Content Assembly



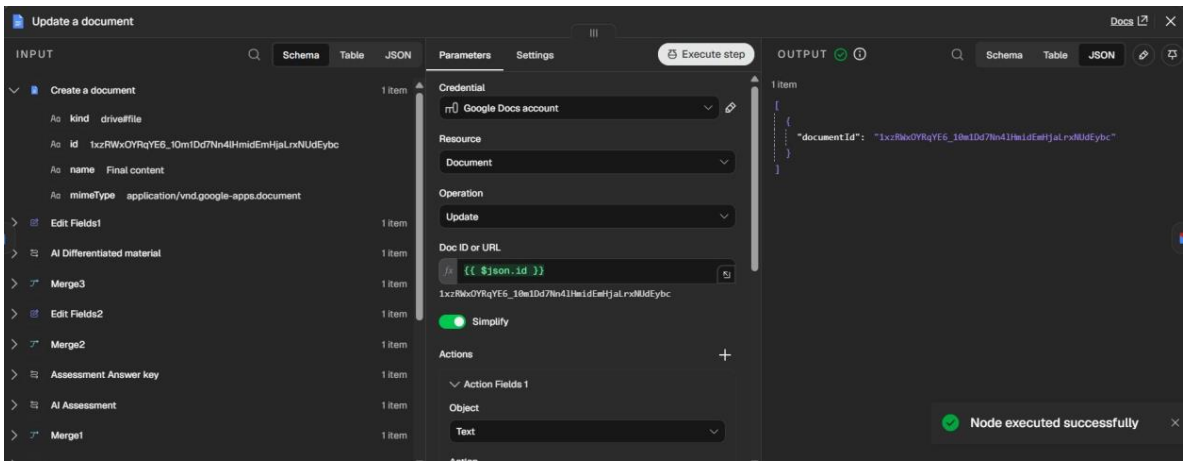
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-14 - Create Document



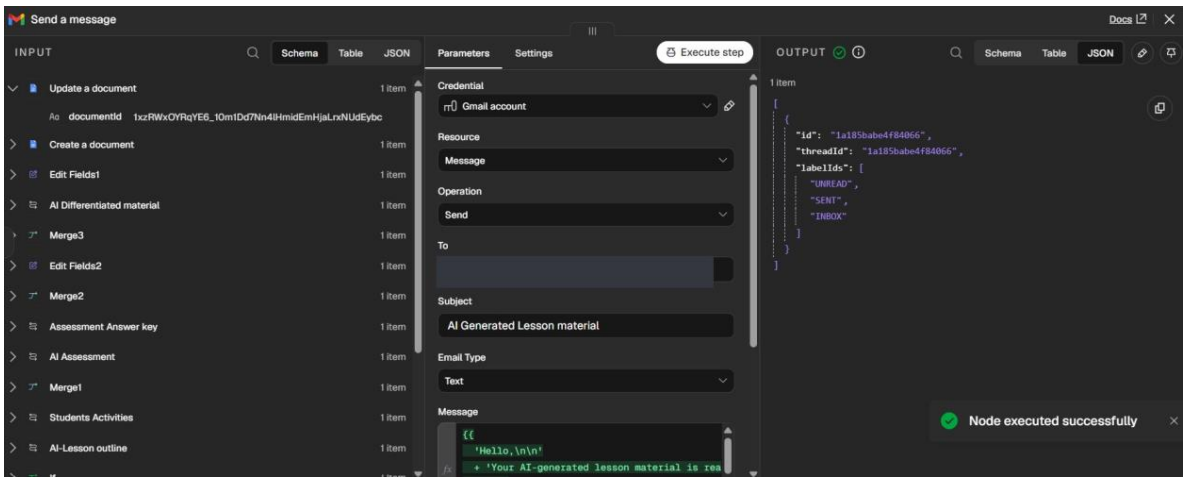
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-15 - Update Document



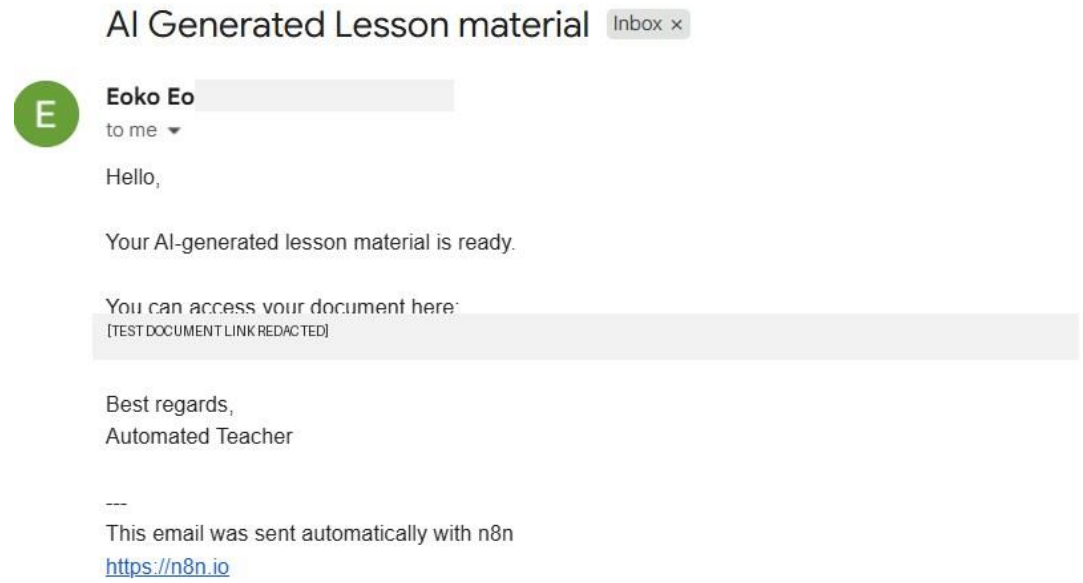
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC16 - Gmail Operation



Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-17 - Email Received



AI Generated Lesson material Inbox ×

E

Eoko Eo

to me

Hello,

Your AI-generated lesson material is ready.

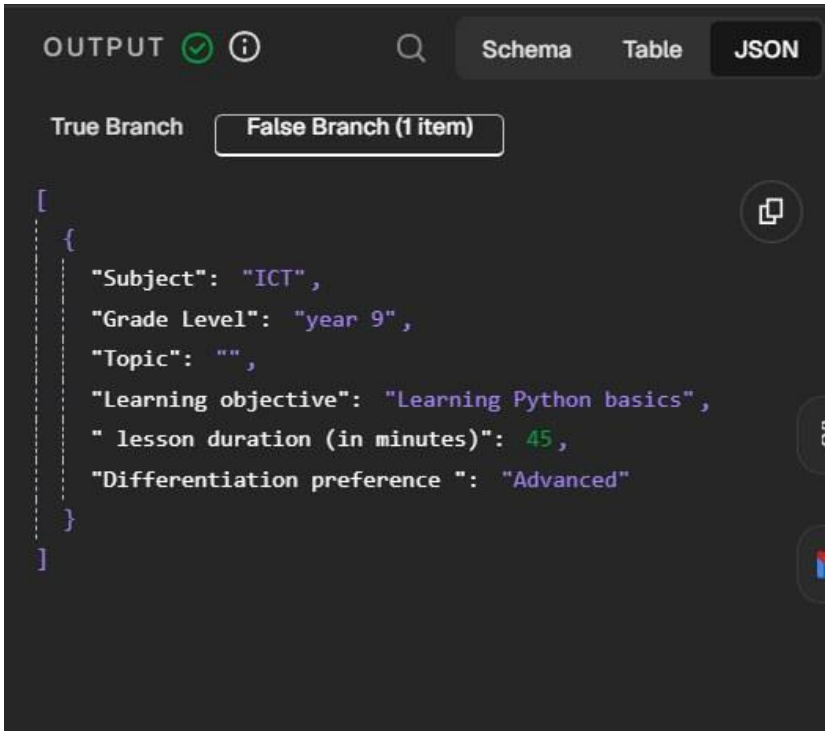
You can access your document here:
[TEST DOCUMENT LINK REDACTED]

Best regards,
Automated Teacher

This email was sent automatically with n8n
https://n8n.io

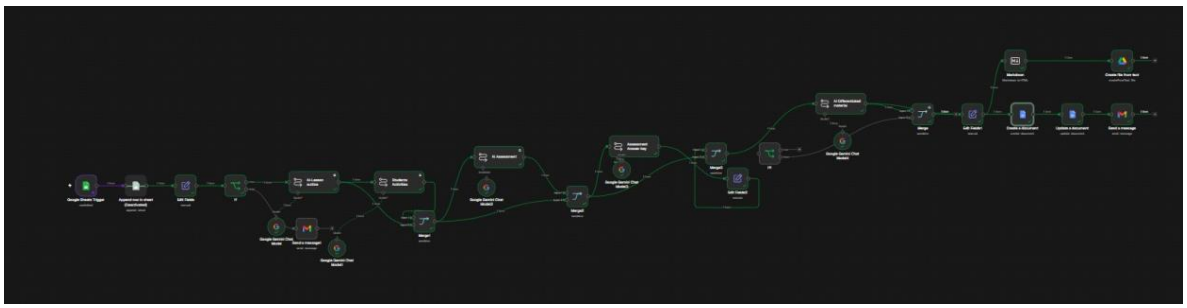
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-18 - If Missing Topic



Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-19 - Integration Execution



Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-20 - Formatted Document

LESSON OUTLINE

Lesson Plan: Introduction to Python Programming

Subject: ICT
Grade Level: Year 9
Topic: Python Basics (Variables, Data Types, and Logic)
Duration: 45 Minutes
Differentiation: Advanced (Focus on procedural abstraction and algorithmic problem solving)

---

1. Starter: Logical Thinking Challenge (5 Minutes)

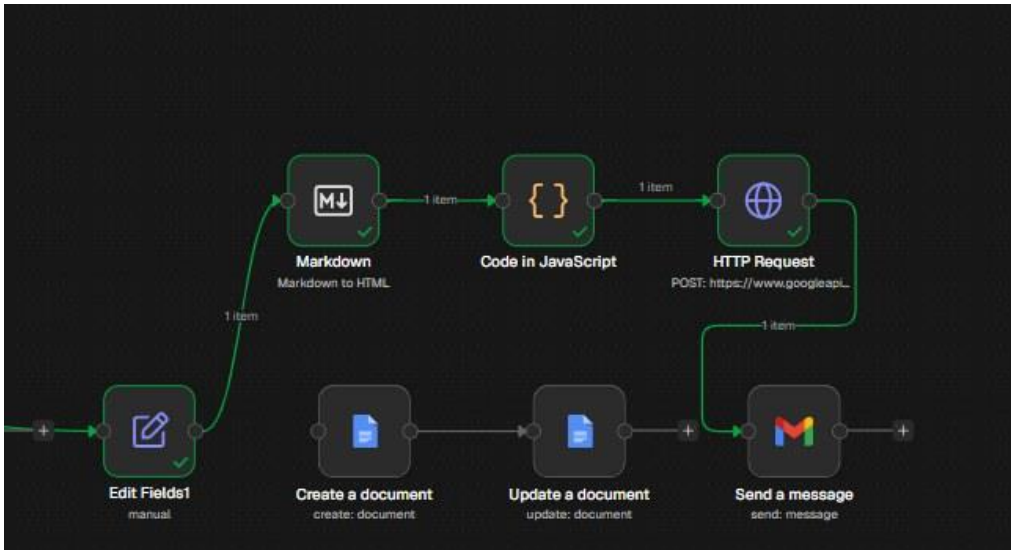
- **Activity:** Present a "pseudocode" snippet on the board representing a common algorithm (e.g., swapping two variables or a simple loop).
- **Task:** Students must predict the output without executing the code.
- **Goal:** Activate prior knowledge of computational logic.

2. Direct Instruction: Core Concepts (10 Minutes)

- **Topics:**
  - Dynamic vs. Static typing in Python.
  - Input/Output handling (input(), print()).
  - Type casting (int(), str(), float()).
  - Introduction to Boolean logic (if, elif, else).

Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-21 - New Final Connections



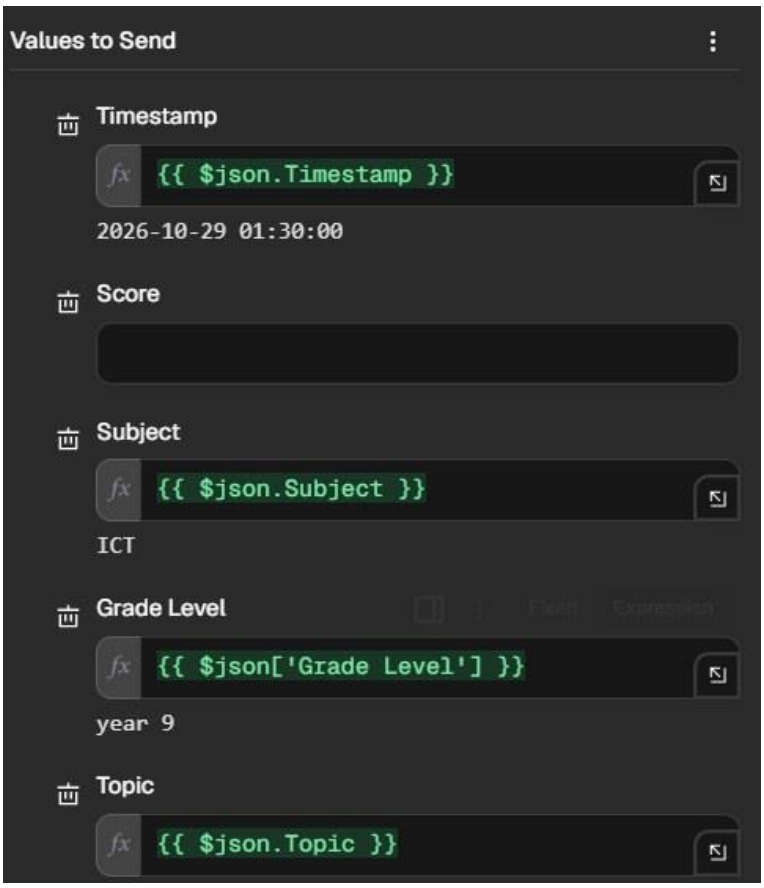
Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-22 - Missing Information Email Output



Source: screenshot from manual workflow testing. Redactions applied where needed.

TC-23 - Append Mapping To Test



Source: screenshot from manual workflow testing. Redactions applied where needed.